

Course Code: CS2403		
Name of the Programme: B.Tech (Hons.) CSE		
Name of the Course: Computer Networks		
Semester: III		
Course credit :	No. of hours per week (L+T+P):	Total no. of Teaching hours:
3	2:0:2	30L+30P = 60 hours
Course Objectives:		
a) Learn about distance-based classification of networks and various mobile communication technologies in the Networking domain		
b) Analyze the system requirements of the Internet and its design parameters for supporting different types of applications		
c) Demonstrate the role of Spanning Tree Protocol in removing loops within a LAN		
d) Understand core Internet protocols (TCP, UDP) and the principles behind VLAN, VPN, NAT, and VoIP.		

Course outline (Syllabus of the course)

Syllabus:

Module - 1: Brief History and Introduction	No. of hours
	6
Brief history of Internet and its evolution; difference between asynchronous and synchronous communication with RS-232 and I2C examples; client-server and peer-to-peer models; distance-based classification of networks, such as PAN, LAN, MAN, WAN, SAN; Introduction to Internet access infrastructure in India. Introduction to the mobile communication technology evolution from 1G to 5G. Node devices, connecting devices and Network Devices(Router, Switch, Bridge, Hub, Gateways)	
Module - 2: Layered Architecture	No. of hours
	6
System requirements of the Internet; network performance parameters, such as bandwidth, latency, delay, jitter, network byte order, generic network messages and headers format, layered architecture, interfaces between the layers and services provided and also introduction to 7 layers of OSI model, TCP/IP Protocol suite.	
Module - 3: Wired/Wireless communications	No. of hours
	4

Ethernet standard 802.3, CSMA/CD, Ethernet frame structure, MAC address format, wireless communications 802.11 frame structure, Wi-Fi, ISM, Bluetooth and piconet.; Three types of switching technologies, datagrams, virtual circuits, source routing.

Module - 4: Addressing Schemes	No. of hours
	8
Internetwork, Internet Protocol (IP), IP service model, IP datagram delivery model; IPv4 header format, need for fragmentation and reassembly and its implementation. Introduction and motivation for IPv6. IPv4 addressing schemes: classful and classless addressing schemes, classful addressing schemes, class A, B, C, D and E, Address Resolution Protocol (ARP); Dynamic Host Configuration Protocol (DHCP), Internet Control Message Protocol (ICMP). Network administration tools, ping, tracert. Motivation behind subnetting and classless addressing.	

Module - 5: Communication Protocols	No. of hours
	6
UDP Protocol, UDP header, Transmission Control Protocol (TCP), TCP header and TCP Vs UDP, Routing Information Protocol (RIP), VLAN, VPN, NAT and VoIP	

Course Outcomes: After completing the course, the students will be able to:	
a.	Elaborate distance-based classification of networks and various mobile communication technologies in the Networking domain
b.	Analyse the system requirements of Internet and its design parameters for supporting different types of applications
c.	Demonstrate the role of Spanning Tree Protocol in removing loops within a LAN
d.	Comprehend and explain IPv4 addressing schemes, routing protocols, and core networking technologies including TCP, UDP, VLAN, VPN, NAT, and VoIP.

Text Books	
1	Larry L. Peterson and Bruce S. Davie., Computer Networks a Systems Approach (The Morgan Kaufmann Series in Networking)- Morgan Kaufmann (2021) 6th ed. ISBN:- 978-0123850591
2	W. Stallings, Data and Computer Communications, 10th Edition, Pearson Education, 2013.

Reference books	
1	Kurose and Ross., Computer Networking, A Top-Down Approach, PEARSON (2013) 7th ed. . ISBN:- 978-0133594140
2	Kozierok M. Charles., TCP/IP Guide, A Comprehensive, Illustrated Internet Protocols Reference, No starch press.
3	Tanenbaum and Wetherall., Computer Networks, Prentice Hall (2011), 5th ed.

Lab Programs		[30 Hours]
PART A- SOFTWARE LAB (Using CISCO Packet tracer)		
1	Construct a network using hub, switches, and routers with hosts and test them using ping.	
2	Connect two different networks using a router with two interfaces a. Ping within the same network b. Ping between different networks	
3	Construct a network with hosts, switches, sniffers to understand ARP protocol and observe ICMP packets using sniffers.	
4	Understand and construct a network with DHCP servers and five routers using DHCP protocol.	
PART B- HARDWARE LAB		
5	IN SWITCH MODE- Introduction to hardware lab by using Telnet and simple ping between the same networks.	
6	IN SWITCH MODE- Construct a VLAN network where communication between same VLAN should be possible and different VLANs should not be possible.	
7.	IN SWITCH MODE- Demonstrate Access Control List in hardware lab.	
8	IN ROUTER MODE- Construct an Inter-VLAN network and demonstrate communication between different VLANs.	
9	IN ROUTER MODE-Configure Router on a Stick configuration using the Hardware lab.	
10	IN ROUTER MODE- Demonstrate the working of OSPF.	

